

Run 05 | OME

India

IV only

Target Opportunity-motivated entrepreneurship; binary 1 vs. 0
Sample 27,555 respondents **Coverage** 1 country; 2015–2022
Positive cases 2,167 (7.86%) **Case time** 52s
Train / val. / test 18,739 / 3,306 / 5,510 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.

Legatum predictors (13): Prosperity index score, Safety and security, Personal freedom, Governance, Social capital, Investment environment, Enterprise conditions, Market access and infrastructure, Economic quality, Living conditions, Health, Education, Natural environment. Total independent variables: 26.

Controls (0): None. Country and respondent ID are excluded as predictors.

Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: CatBoost (0.9874). Highest F1: XGBoost (0.7947). The all-negative accuracy baseline is 92.14% (full sample).

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
CatBoost	0.9874	0.8477	96.32	70.91	90.07	0.7935	11s
Random Forest	0.9873	0.8538	96.24	71.08	87.99	0.7864	1s
H2O GBM	0.9872	0.8446	96.03	73.06	78.29	0.7559	3s
XGBoost	0.9869	0.8389	96.35	71.25	89.84	0.7947	<1s
Elastic Net	0.9860	0.8343	96.03	72.11	80.60	0.7612	11s
Logistic Regression	0.9858	0.8326	95.97	71.49	81.06	0.7597	1s
Decision Tree	0.9858	0.8121	96.32	75.00	79.68	0.7727	<1s
Lasso	0.9857	0.8255	96.04	71.81	81.76	0.7646	8s
Ridge	0.9854	0.8256	95.92	71.58	79.68	0.7541	10s
Naive Bayes	0.9046	0.5639	92.99	54.53	65.36	0.5945	5s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	792.30	47.18
2	Established business owner	271.26	16.15
3	Business angel	83.22	4.96
4	Knowing entrepreneurs	70.07	4.17
5	Business exit	43.22	2.57
6	Self-efficacy	25.43	1.51
7	Good career choice	23.90	1.42
8	Social enterprise	21.73	1.29

Share is the percentage of total forest impurity importance across all 51 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Legatum predictors within the same forest

Rank	Legatum predictor	Raw importance	Share%
1	Social capital	13.73	0.82
2	Investment environment	13.39	0.80
3	Prosperity index score	12.39	0.74

Legatum subset ranked by raw importance; shares use total forest importance, not the Legatum subtotal.

Source: Completed Objective 2 batch, 28 September 2026; 05_obj2_india_iv_OME. Scores and importance are taken directly from saved CSV outputs.